

Kumar Ashutosh

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🌐 <https://ashutoshkr.me>

Education

- 2021 – 2026 📖 **The University of Texas at Austin** *Austin, TX*
Ph.D. in Computer Science
Advisor: Kristen Grauman
Thesis title: *Video Understanding for Learning and Assessing Skill*
- 2016 – 2021 📖 **Indian Institute of Technology Bombay** *Mumbai, India*
Master's degree in Electrical Engineering
Thesis title: *3D Shape Reconstruction with View-Planning*.
- 📖 **Indian Institute of Technology Bombay**
Bachelor's degree in Electrical Engineering with a Minor in Computer Science

Work Experience

- 2026-Present 📖 **Research Scientist**, Meta Reality Labs *Sunnyvale, CA*
Building AI-coach algorithms for Aria AI glasses 🧐 that perceive human activity and guide users in learning real-world skills.

Prior Internships and Research Positions

- 2021-26 📖 **UT Austin**, Graduate Research Assistant *Austin, TX*
- Summer 2025 📖 **GenAI, Meta**, Research Scientist Intern *New York, NY*
- 2023-25 📖 **FAIR, Meta**, Visiting Researcher *Austin, TX*
- Summer 2022 📖 **FAIR, Meta**, Research Intern *New York, NY*
- Winter 2019 📖 **360World**, AR/VR Developer Intern *Budapest, Hungary*
- Summer 2019 📖 **Sony Corporation**, Research Engineer Intern *Kanagawa, Japan*
- Summer 2018 📖 **National University of Singapore**, Research Intern *Singapore*
- 📖 **Google Summer of Code**, Developer *Remote*

Research Publications

Preprints

- 1 **K. Ashutosh**, C. H. Wu, and K. Grauman, "SportSkills: Physical Skill Learning from Sports Instructional Videos," Mar. 2026. arXiv: 2603.25163 [cs.CV].
- 2 **K. Ashutosh**, X. Wang, X. Yin, *et al.*, "Human detectors are surprisingly powerful reward models," Jan. 2026. arXiv: 2601.14037 [cs.CV].
- 3 **K. Ashutosh**, R. Girdhar, L. Torresani, and K. Grauman, "What You Say Is What You Show: Visual Narration Detection in Instructional Videos," 2023. arXiv: 2301.02307 [cs.CV].

Conference Proceedings

- 1 C. H. Wu, **K. Ashutosh**, and K. Grauman, "SkillSight: Efficient First-Person Skill Assessment with Gaze," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026, 🧐.

- 2 C. H. Wu*, **K. Ashutosh***, and K. Grauman, "Stitch-a-Demo: Video Demonstration from Multistep Descriptions," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026, [🔗](#).
- 3 **K. Ashutosh** and K. Grauman, "Learning Skill-Attributes for Transferable Assessment in Video," in *Advances in Neural Information Processing Systems (NeurIPS)*, Dec. 2025, [🔗](#).
- 4 M. Huh, Z. Xue, U. Das, **K. Ashutosh**, K. Grauman, and A. Pavel, "Vid2Coach: Transforming How-To Videos into Task Assistants," in *The ACM Symposium on User Interface Software and Technology (UIST)*, Sep. 2025, [🔗](#).
- 5 **K. Ashutosh**, Y. Gandelsman, X. Chen, I. Misra, and R. Girdhar, "LLMs can see and hear without any training," in *Proceedings of the International Conference on Machine Learning (ICML)*, Jul. 2025, [🔗](#).
- 6 **K. Ashutosh**, T. Nagarajan, G. Pavlakos, K. Kitani, and K. Grauman, "ExpertAF: Expert Actionable Feedback from Video," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025, [🔗](#).
- 7 **K. Ashutosh**, G. Pavlakos, and K. Grauman, "Fiction: 4D Future Interaction Prediction from Video," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025, **Highlight paper (Top 3%)** [🔗](#).
- 8 **K. Ashutosh**, Z. Xue, T. Nagarajan, and K. Grauman, "Detours for Navigating Instructional Videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024, **Highlight paper (Top 2.8%)** [🔗](#).
- 9 C. Chen, **K. Ashutosh**, R. Girdhar, D. Harwath, and K. Grauman, "SoundingActions: Learning How Actions Sound from Narrated Egocentric Videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024.
- 10 K. Grauman, A. Westbury, L. Torresani, *et al.*, "Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024, **Oral paper (Top 0.8%)** [🔗](#).
- 11 Z. Xue, **K. Ashutosh**, and K. Grauman, "Learning Object State Changes in Videos: An Open-World Perspective," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024, [🔗](#).
- 12 **K. Ashutosh**, S. Ramakrishnan, T. Afouras, and K. Grauman, "Video-Mined Task Graphs for Keystep Recognition in Instructional Videos," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023, [🔗](#).
- 13 **K. Ashutosh**, R. Girdhar, L. Torresani, and K. Grauman, "HierVL: Learning Hierarchical Video-Language Embeddings," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2023, **Highlight paper (Top 2.5%)** [🔗](#).
- 14 A. Jaiswal, **K. Ashutosh**, J. F. Rousseau, Y. Peng, Z. Wang, and Y. Ding, "RoS-KD: A Robust Stochastic Knowledge Distillation Approach for Noisy Medical Imaging," in *2022 IEEE International Conference on Data Mining (ICDM)*, Dec. 2022, [🔗](#).
- 15 **K. Ashutosh**, S. Kumar, and S. Chaudhuri, "3D-NVS: A 3D Supervision Approach for Next View Selection," in *2022 26th International Conference on Pattern Recognition (ICPR)*, Aug. 2022, [🔗](#).
- 16 **K. Ashutosh**, J. Nair, A. Kagrecha, and K. Jagannathan, "Bandit algorithms: Letting go of logarithmic regret for statistical robustness," in *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021, **Oral presentation (Top 3%)** [🔗](#).
- 17 **K. Ashutosh**, "Hardware Performance Analysis of Mobile-Based Augmented Reality Systems," in *2020 International Conference on Computational Performance Evaluation (ComPE)*, 2020, [🔗](#).

- 18 **K. Ashutosh**, S. Consul, B. Dedhia, P. Khirwadkar, S. Shah, and S. Kalyanakrishnan, "Lower Bounds for Policy Iteration on Multi-action MDPs," in *2020 59th IEEE Conference on Decision and Control (CDC)*, 2020, [🔗](#).
- 19 R. Bose, **K. Ashutosh**, J. Li, A. Dragomir, N. Thakor, and A. Bezerianos, "A Multilayer Network Approach for Studying Creative Ideation from EEG," in *Brain Informatics*, Springer International Publishing, 2018, [🔗](#).

Professional Service

Reviewer

CVPR 2024	📌 Outstanding reviewer (Top 2% out of 10k reviewers)
NeurIPS 2024	📌 Top reviewer (Top 8% out of 14k reviewers)
2023-25	📌 The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
	📌 The IEEE/CVF International Conference on Computer Vision (ICCV)
2022-24	📌 European Conference on Computer Vision (ECCV)
2025	📌 International Journal of Computer Vision (IJCV)
	📌 International Conference on Machine Learning (ICML)
2024	📌 Neural Information Processing Systems (NeurIPS)
2024-25, 22	📌 The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
2024	📌 Asian Conference on Computer Vision (ACCV)
2025	📌 AAAI Conference on Artificial Intelligence (AAAI)

Teaching Assistant

Spring 2025	📌 <i>Visual Recognition</i> in Computer Science, UT Austin
Spring 2021	📌 <i>Matrix Computations</i> in Electrical Engineering, IIT Bombay
Autumn 2020	📌 <i>Applied Linear Algebra</i> in Electrical Engineering, IIT Bombay

Responsibilities

2019 – 2020	📌 Department Academic Mentor Coordinator in Electrical Engineering, IIT Bombay
2018 – 2021	📌 Institute and Department Academic Mentor, IIT Bombay
2017 – 2018	📌 Convener of <i>Web n Coding Club</i> , IIT Bombay

Open Source

2025	📌 Open sourced code implementations of authored papers: 🔗 , 🔗 , 🔗 , 🔗 , 🔗 (500+ stars)
2023	📌 Contributed to PyTorch codebase 🔗
2017	📌 Contributed 2k+ lines of code to scikit-learn, a popular ML package 🔗

Awards and Achievements

2026	📌 Egocentric Vision (EgoVis) 2024/2025 Distinguished Paper Award for ExpertAF (CVPR 2025)
	📌 Egocentric Vision (EgoVis) 2024/2025 Distinguished Paper Award for Ego-Exo4D (CVPR 2024)
2025	📌 Egocentric Vision (EgoVis) 2023/2024 Distinguished Paper Award for HierVL (CVPR 2023)
	📌 Professional Development Award by UT Austin to attend CVPR 2025
2023	📌 Professional Development Award by UT Austin to attend NeurIPS 2023

Awards and Achievements (continued)

- 2020  Professional Development Award by UT Austin to attend CVPR 2023
- 2020  Department Color by IIT Bombay for valuable contribution to the mentorship program
- 2017  Invited to the Republic Day Parade as a guest of the Hon'ble Prime Minister of India
- 2016  Rashtrapati Puraskar (President's Award) by the Hon'ble President of India for Scouting
-  99.97 (out of 1.2M) and 99.14 percentile (out of 0.15M) in JEE Mains and Advanced
- 2015  Qualified Indian National Mathematical Olympiad and attended selection camp for IMO
-  Cleared NTSE and KVPY scholarship exams organized by the Govt. of India

Press Coverage

- 2025  Blogs and discussions of MILS (ICML 2025) by readers:  (Medium)  (YCombinator News)
- 2024  Ego-Exo4D's (CVPR 2024) media coverage:  
- 2023  Meta AI's coverage of HierVL (CVPR 2023) paper : , , , 

Talks

- 2025  Oral presentation of Fiction (CVPR 2025) at Egocentric Vision (EgoVis) Workshop, CVPR 2025
-  Invited talk to present our paper MILS (ICML 2025) at Cohere AI
- 2024  Invited talk to present our paper ExpertAF (CVPR 2025) at Ego4D winter symposium 2024
-  Invited talk at IIT Delhi in Jan titled "Long-video understanding with text supervision"
- 2023  Highlight presentation of HierVL at CVPR 2023 
-  Invited talk at International Workshop on Large Scale Holistic Video Understanding, CVPR 2023
- 2021  Oral presentation of our work at AISTATS 2021 
- 2020  Invited talk on Augmented Reality Applications at Electronics and Robotics Club, IIT Bombay

Technical Skills

- Languages  Python, C++, C, HTML, CSS, $\text{\LaTeX}$
- ML Tools  Huggingface, Deepspeed, Fairseq, SLURM, PyTorch, Tensorflow, Git
- Development  React, Jekyll, Android Studio, Xcode, Unity

Extracurricular Activities

- Sports  Football (Soccer), Squash, Cricket, Tennis, Badminton
- Activities  Hiking, Running
- Music  Guitar, Piano/Keyboard